

Phase-II

Solution Requirements (Functional & Non-functional)

Date	02 March 2026
Team ID	Team 154
Project Name	Insurance Fraud Detection using ML
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Preparation	Cleaning missing values and handling '?' entries.
		Outlier treatment using IQR and Log transformations.
		Feature encoding (Label/One-Hot) and scaling (StandardScaler).
FR-2	Model Development	Training 12 different classification algorithms for comparison.
		Cross-validation and hyperparameter optimization.
		Selecting the best-performing model based on accuracy/recall.
FR-3	Model Serialization	Saving the trained model and scaler using joblib.

		Exporting feature columns to ensure input consistency.
FR-4	Deployment & Prediction	Developing a Flask-based web interface for user input.
		Real-time prediction service to flag "Fraud" or "Legal" claims.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The web interface must be intuitive, allowing investigators to enter claim data easily without technical training.
NFR-2	Security	Sensitive insurance and policyholder data must be handled securely during the prediction process.
NFR-3	Reliability	The model must provide consistent and reproducible predictions across various claim scenarios.
NFR-4	Performance	The application must return fraud predictions in under 1 second to ensure an efficient workflow.
NFR-5	Availability	The system should be accessible through a web browser 24/7 for insurance adjusters.

NFR-6	Scalability	The architecture should be capable of processing an increasing volume of daily claims as the dataset grows.
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